

Open Source Smart Espresso Machine

TED Talk, 3/6/26

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Who are we again?

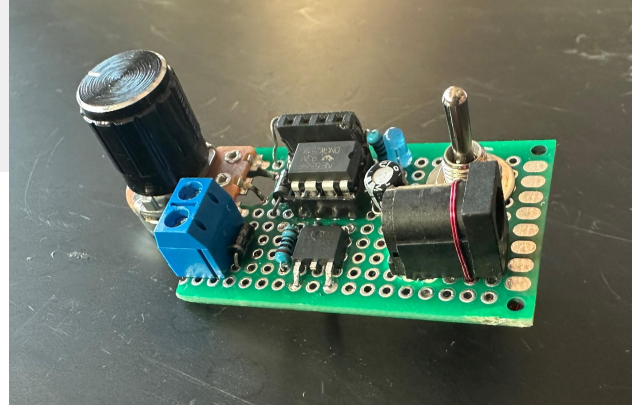
- PSU **OsSEM**, Open-source Smart Espresso Machine
- Sponsored by **Healthy Buildings Research** and **Coffee & Telesensing Laboratories**
- Functional enhancement modification for Gaggia Classic espresso machine
 - powered by STM32 and ESP32 hardware
 - open-source software/firmware installations



Updates:

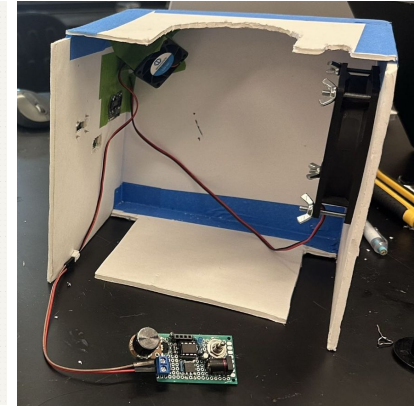
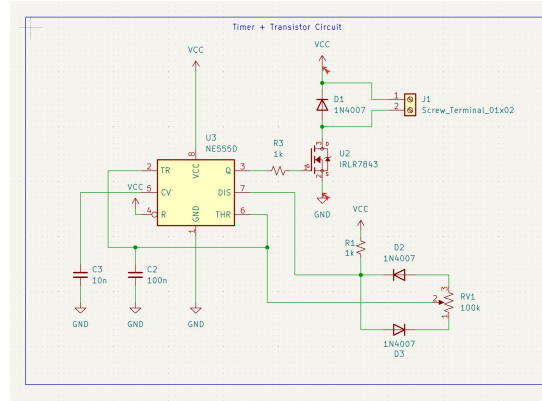
- **ECE:**

- Adjustable fan speed controller
- Initial measurements
- Full sensor integration
- Block diagrams and other documentation



- **ME:**

- 3D printed enclosure
- CNC machining for enclosure rev 3
- Machined bottomless Portafilter



Updates:

- **What's going great:**
 - Initial measurements
- **What's going not-so-great:**
 - Waiting on parts
 - Other course projects/labs/linals
- **What we wish we knew:**
 - Relative concentrations of released compounds (how many ppm or ppb?)
- **Surprises?**
 - How fast time flies by

